Q03 — Deeper Projects

Is reasoning real, a property of language, a reflection, or an illusion?

DLS 2026 — AI Internals Module

Framing the Question

When a large language model emits a chain of intermediate tokens before an answer, we are tempted to call this "reasoning." But the philosophical content of that word is contested. Four hypotheses structure this module: (i) *reasoning is real*—there is a causal, generalizable computation that mediates premises and conclusions; (ii) *reasoning is a property of language*—the surface form of trained text induces inference-like patterns without a separable mechanism; (iii) *reasoning is a reflection*—the model mirrors statistical regularities of human-produced argument without executing the argument; (iv) *reasoning is an illusion*—chain-of-thought (CoT) text is post-hoc rationalization that does not reflect the model's actual decision process. These are not mutually exclusive across tasks, and the empirical literature supports a nuanced view: CoT improves performance on some tasks [1, 2], yet verbalized explanations can be unfaithful to the underlying computation [3, 4], and apparent reasoning degrades under distribution shift and token-probability sensitivity [5]. The projects below ask you to generate *evidence*, not opinions.

1 Project A — Faithfulness Auditing of Chain-of-Thought

Goal. Determine whether verbalized reasoning steps are causally responsible for a model's answer, or whether they are decorative.

Method. Reproduce and extend the biasing and perturbation protocols of **(author?)** [3] and **(author?)** [4]. Construct matched prompt pairs in which a spurious cue (e.g., answer order, few-shot label bias) is silently inserted. Measure whether the model's answer flips while its stated rationale fails to mention the cue. Then run causal interventions: truncate CoT, paraphrase it, inject errors into intermediate steps, and measure answer sensitivity.

Deliverables.

- A faithfulness benchmark of ≥ 400 items across ≥ 3 task families (arithmetic, multi-hop QA, logical entailment), with per-item bias labels.
- A quantitative report giving: answer-flip rate under biasing, rationale-mention rate, and an *early-answering* / *added-mistake* sensitivity curve per **(author?)** [4].
- A written argument (3–4 pp.) assigning each task family to one of the four hypotheses, with confidence intervals.

2 Project B — Mechanistic Localization of an Inference Circuit

Goal. Test whether a specific reasoning behavior corresponds to an identifiable, reusable internal computation (the "reasoning is real" hypothesis at the mechanism level).

Method. Choose a narrow, well-posed task (e.g., multi-step in-context induction, or a bracketed arithmetic subtask). Apply activation patching and path patching following **(author?)** [6] and the causal mediation framework of **(author?)** [7]. Use attention-pattern analysis and logit attribution to identify candidate circuits, then validate by ablation and transfer to held-out instances. Document whether the circuit generalizes or is template-bound (evidence for "property of language").

Deliverables.

- A circuit diagram with the heads/MLPs implicated, plus patching effect sizes.
- A transfer table: circuit intact across ≥ 4 structural variants (vocabulary swap, length scaling, distractors).
- A short claim with falsification criterion: state what observation would refute "this circuit implements the inference."

3 Project C — Robustness and the "Embers" of Autoregression

Goal. Quantify how much apparent reasoning is governed by token-level probability rather than task logic.

Method. Following **(author?)** [5], construct task variants that hold logical structure fixed while varying output probability (common vs. rare target strings, common vs. rare task framings). Measure accuracy as a function of estimated output rarity and task frequency. Contrast with symbolic baselines. Optionally cross-reference counting/length-sensitivity failures reported in compositional-generalization work [8].

Deliverables.

- Accuracy-vs-rarity curves with regression coefficients isolating the probability effect from the logic effect.
- A "reflection index": the fraction of variance in correctness explained by output probability versus task structure.
- Interpretation mapping results onto the four hypotheses.

4 Project D — Operationalizing the Definition

Goal. Rather than settle the metaphysics, propose a *measurable* definition of reasoning and defend it.

Method. Draft operational criteria (e.g., causal dependence of answer on verbalized steps, out-of-distribution generalization, circuit reusability, invariance to superficial rephrasing). Relate your criteria to emergent-capability debates [9, 10] and to the self-consistency improvements of **(author?)** [11]. Apply your criteria to results from Projects A–C.

Deliverables.

- A 2-page definitional proposal with explicit metrics and thresholds.
- A scoring of ≥ 2 models against your criteria, with a verdict per hypothesis (real / linguistic / reflective / illusory), per task.
- A critical appendix noting where your own definition is itself at risk of measuring an illusion.

Assessment

Projects are graded on experimental validity, honesty about negative results, and the tightness of the link between *evidence* and *claim*. A project that convincingly demonstrates "illusion" on a task earns the same credit as one demonstrating "real" reasoning; intellectual direction is not rewarded, only rigor.

References

[1] J. Wei, X. Wang, D. Schuurmans, M. Bosma, B. Ichter, F. Xia, E. Chi, Q. Le, and D. Zhou. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

[2] M. Nye, A. J. Andreassen, G. Gur-Ari, H. Michalewski, J. Austin, D. Bieber, D. Dohan, A. Lewkowycz, M. Bosma, D. Luan, C. Sutton, and A. Odena. Show your work: Scratchpads for intermediate computation with language models. *arXiv preprint arXiv:2112.00114*, 2021.

[3] M. Turpin, J. Michael, E. Perez, and S. R. Bowman. Language models don't always say what they think: Unfaithful explanations in chain-of-thought prompting. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[4] T. Lanham, A. Chen, A. Radhakrishnan, B. Steiner, C. Denison, D. Hernandez, D. Li, E. Durmus, E. Hubinger, J. Kernion, et al. Measuring faithfulness in chain-of-thought reasoning. *arXiv preprint arXiv:2307.13702*, 2023.

[5] R. T. McCoy, S. Yao, D. Friedman, M. Hardy, and T. L. Griffiths. Embers of autoregression: Understanding large language models through the problem they are trained to solve. *arXiv preprint arXiv:2309.13638*, 2023.

[6] K. Wang, A. Variengien, A. Conmy, B. Shlegeris, and J. Steinhardt. Interpretability in the wild: A circuit for indirect object identification in GPT-2 small. In *International Conference on Learning Representations (ICLR)*, 2023.

[7] K. Meng, D. Bau, A. Andonian, and Y. Belinkov. Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

[8] B. M. Lake and M. Baroni. Generalization without systematicity: On the compositional skills of sequence-to-sequence recurrent networks. In *International Conference on Machine Learning (ICML)*, 2018.

[9] J. Wei, Y. Tay, R. Bommasani, C. Raffel, B. Zoph, S. Borgeaud, D. Yogatama, M. Bosma, D. Zhou, D. Metzler, et al. Emergent abilities of large language models. *Transactions on Machine Learning Research (TMLR)*, 2022.

[10] R. Schaeffer, B. Miranda, and S. Koyejo. Are emergent abilities of large language models a mirage? In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[11] X. Wang, J. Wei, D. Schuurmans, Q. Le, E. Chi, S. Narang, A. Chowdhery, and D. Zhou. Self-consistency improves chain of thought reasoning in language models. In *International Conference on Learning Representations (ICLR)*, 2023.